SAINT FRANCIS UNIVERSITY
Loretto, Pennsylvania
INTERNSHIP CONTRACT

I. **Student Name:** Brennen Yingling **Major:** Computer Science / Cybersecurity

II. Briefly describe your intended internship, noting course work and experience that will enable you to profit from this particular internship listing the duties of the internship. *(One-half page minimum.)*

The proposed six-week Summer Research Program includes the hardware-in-the-loop (HITL) engineering, software refactoring, and payload integration of Flatsat satellite diagnostic system. Previous course work in object-oriented programming (OOP) and digital systems design gives the required technical knowledge that will allow myself to fully capitalize on this specialized internship opportunity. The experience with Python programming language will directly translate into modular optimization of the flight software, while lab familiarity with electronics, wiring diagrams, and digital multimeters test equipment will enable me to make an instant contribution to electrical debugging and components integration.

According to operational logs, the key responsibilities of this internship program include full lifecycle aerospace systems engineering tasks. The technical responsibility of the role entails structural wire harness construction including precision soldering of 9-pin, 15-pin, and 25-pin D-sub connectors to power supply boards, switchboards, and GPIO Pi Hats. The software responsibilities involve refactoring the monolithic flight code to modular class-based object architectures to support system diagnostics, ESim heartbeat monitoring, and USB-over-Ethernet commanding loops. Moreover, the job will require HITL troubleshooting to overcome power bus constraints, floating grounds, and insertion losses in the process of SDR payload, X-rays sensors, and Arducam Starvis IMX585 optical sensor integration.

In summary, this internship serves as a means by which theory and aerospace application are linked directly. From developing hardware to implementing the software, this internship is a perfect match to my career path as a software engineer. I am more than willing to utilize my academic background in solving such technical issues and guarantee the reliability of the Flatsat while increasing my engineering skills.

III. Location of Internship: (include site name, city and state)

Saint Francis University, Loretto PA

If the internship location is in the State of Colorado, has workers' compensation insurance coverage been arranged as is required? ____ Yes ____ No **X** N/A
(Attach copy of Certificate of Insurance.)

Visit http://francis.edu/state-authorization/ for information on State of Colorado requirements and instructions for obtaining workers' compensation coverage.

If the internship location is in the State of New Hampshire, has the program and worksite been approved by the New Hampshire Department of Labor? ___ Yes ___ No **X** N/A

IV. Out-of-state and international internships are only possible in states and countries where Saint Francis University has authorization. Does Saint Francis University have authorization for internships in the state or country of the proposed internship location? **X** Yes ____ No

Visit http://francis.edu/state-authorization/ to answer the question or contact your advisor or internship faculty supervisor, who can collaborate with the Academic Compliance Specialist.

V. Some internship sites require a written agreement between the University and the internship site. Does a written agreement exist between the University and internship site for this experience?
___ Yes **X** No
If unsure, consult your academic department internship coordinator.

VI. On-Site Supervisor: **Dr. Lanika Ruzhitskaya**

Saint Francis University Faculty Supervisor: **Dr. Kevin Slonka**

VII. Credits to be earned/course number(s): **3 credits**
(example: 3 credits: COMM 398 – 1 credit; COMM 399 – 2 credits)

VIII. Starting/Ending dates of internship: From: 8/24/26 To: 12/11/2026

(over)

IX. Title/Author of book you plan to read in connection with your internship:

The Pheonix Project by Gene Kim

X. Please state three educational objectives that you hope to accomplish. **Be precise.**

1. I hope to better my programming skills in the Python language.
2. I hope to gain experience programming in a team and gain a better understanding of Git.
3. I hope to better my troubleshooting skills.

XI. Describe the three credit hour project that you will complete during the course of this internship. (One page minimum.)

As part of this six weeks internship, the key task that was assigned to me for my project of three credit hours was to design and construct an entire FlatSat diagnostic system. My FlatSat was a flat representation of our satellite's bus and payload electronic systems on the lab desk. The advantage of such layout provided me with an ability to test hardware in loop (HITL), validate software and debug all the electrical circuits prior to putting everything together inside of the tightly packed CubeSat frame. The ultimate goal of my project was to build a fully functional prototype allowing integration of our flight hardware with custom object-oriented software.

The first stage of my project was largely dominated by electrical engineering and the fabrication of the avionics wire harnesses. The main objective of my project was to design a specialized and customized soldering of power and data lines for interfacing our three different power sources, hardware switch board, Raspberry Pi 4 micro controller and our set of mission payloads. For this purpose, I had to interpret engineering pinouts in the form of actual wiring. Fabrication of these circuits entailed accurate soldering of 22-gauge and 24-gauge wires in 9-pin, 15-pin and 25-pin D-sub connectors to create data channels, like the 25-pin D-sub to Raspberry Pi 4 GPIO Pi Hat.

In line with the hardware assembly process, I undertook an extensive flight software development and refactoring process using the Python programming language. In order to manage our system well, a critical software component of my project required refactoring of a large, bulky 1,000-line codebase into a neat, object-oriented design. Breaking down our system into manageable classes within 100 lines each gave me the ability to easily track our parameters. My software design is characterized by an all-encompassing Mode Manager program that incorporates unique states such as the Idle state which initiates system health checks, data file loading, ESim heartbeat, and USB-over-Ethernet commanding loops.

The final step in my project involved thorough hardware-in-the-loop payload integration and environmental troubleshooting. The integration process of our important mission payloads such as the SDR for generating .IQ files, X-Ray sensor and Arducam Starvis IMX585 optical sensor was done in a systematic manner within the switchboard topology. Through the use of a digital multimeter, I was able to detect complicated problems that were encountered during the engineering process such as floating grounds on unoccupied pins, poor soldering joints and drastic drop in bus voltages, and eventually solved them using customized wiring and topology modifications.

Brennen Yingling 7/18/2026

Student/Date

Lanika Ruzhitskaya

On-Site Supervisor/Date

07/21/2026

Saint Francis University Supervisor/Date

Department Chair/Date

Forward completed form to Career Services (Scotus 323).
Received by Career Services _____date _____ initial.
Received by Registrar's Office _____date _____ initial.